

Reducing Medication Errors Through a Digital Medication Verification System

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I. Introduction

Medication administration is one of the important responsibilities of nurses because it directly affects patient safety and quality of care. Nurses are expected to prepare, verify, administer, and document medications accurately while caring for patients with different conditions and needs. However, medication errors can occur at different stages of the medication-use process. The World Health Organization (WHO) identifies unsafe medication practices and medication errors as a leading cause of avoidable harm in health care systems worldwide. WHO also estimates that the global cost associated with medication errors is about US\$42 billion annually.

In actual nursing environments, medication administration may be affected by factors such as interruptions, workload, fatigue, poor working conditions, incorrect patient identification, unclear medication information, and documentation problems. These factors can make it more difficult for healthcare workers to consistently perform all required safety checks. Because nurses are closely involved in bedside medication administration, an information-technology solution can provide an additional verification layer without removing the nurse's professional responsibility.

This proposal presents a Digital Medication Verification System based on barcode-assisted medication administration. The proposed system connects the patient's identification, the medication barcode, and the electronic medication order. Its purpose is to help nurses identify mismatches before medication administration and to support accurate electronic documentation. The project focuses on how technology can strengthen established nursing medication-safety practices.

Problem description

Medication errors remain a significant patient-safety concern because mistakes can happen during prescribing, transcribing, dispensing, preparing, administering, and monitoring medications. In nursing practice, the administration stage requires careful verification of the patient, medication, dose, route, and time. Even when nurses follow established procedures, human and system factors can create opportunities for error.

The problem becomes more challenging in busy clinical settings where nurses may care for several patients and perform multiple tasks within a limited period. Interruptions, heavy workloads, fatigue, incomplete information, unreadable labels, and documentation demands may interfere with a consistent medication-checking process. WHO notes that human factors such as fatigue, poor environmental conditions, and staff shortages can contribute to medication errors.

Barcode-assisted medication administration (BCMA) was developed to provide an electronic safety check by scanning machine-readable identifiers on the patient and medication and connecting these checks with an electronic medication administration record. Research has shown that BCMA can identify medication-error categories that may not be detected through traditional checking alone. However, evidence about its effect is not completely consistent. A 2026 systematic review found weak-to-moderate evidence that BCMA can lower some categories of medication-administration errors in non-unit-dose settings, while noting that more research is needed.

There are also implementation concerns. Studies have reported workarounds related to unreadable barcodes, malfunctioning scanners, missing wristbands, connectivity problems, and workflow limitations. Therefore, the problem is not simply the absence of technology. The proposed system must be designed around real nursing workflows, supported by training, and used together with professional judgment and existing medication-safety procedures.

Proposal solution.

The proposed solution is the development of a Digital Medication Verification System using Barcode-Assisted Medication Administration (BCMA) integrated with an electronic Medication Administration Record (eMAR). The system is designed to support nurses by connecting patient identification, medication information, electronic medication orders, alerts, and documentation in one workflow.

Before administering a medication, the nurse first scans the patient's identification barcode, such as the barcode on the patient's identification wristband. The nurse then scans the medication barcode. The system compares the scanned information with the patient's electronic medication order. If the information matches, the system allows the nurse to continue with the medication-administration process. If a mismatch is detected, the system produces an alert so that the nurse can stop and investigate the discrepancy before administration.

The system also supports electronic documentation. After a medication is properly verified and administered, the administration information can be recorded in the electronic medication administration record. This reduces repetitive manual recording and makes medication information easier for authorized healthcare personnel to access. The system can also provide records that may be used for quality-improvement monitoring.

The main users of the proposed system are registered nurses and other authorized healthcare professionals involved in medication administration. Nursing students may use the system only within their clinical scope and under appropriate supervision. The system may be implemented in hospitals, clinics, and other healthcare settings where electronic medication records and barcode technology can be appropriately integrated.

Expected benefits include improved patient identification, additional medication-verification checks, faster and more consistent documentation, earlier identification of mismatches, and stronger support for nursing patient-safety practices. Nevertheless, the system should not be treated as a replacement for nursing judgment. Proper training, reliable equipment, appropriate system design, privacy protection, and clear downtime procedures are necessary for safe implementation.

Conclusion

Medication errors are an important patient-safety problem that can affect patients, nurses, and healthcare organizations. They may occur because of weaknesses in medication systems as well as human factors such as fatigue, interruptions, workload, and poor working conditions. Since nurses play a major role in medication administration, supporting their safety checks with appropriate information technology can help strengthen the medication-use process.

This proposal highlights a Digital Medication Verification System using barcode-assisted medication administration and an electronic medication administration record. By scanning the patient and medication and comparing the information with the electronic order, the system can provide an additional check before medication administration. It can also support electronic documentation and create useful information for patient-safety and quality-improvement activities.

However, technology alone cannot eliminate medication errors. Evidence on BCMA effectiveness is supportive but not uniformly conclusive, and implementation problems can create workarounds or other safety risks. For this reason, the proposed system should be introduced together with nursing education, workflow testing, reliable equipment, system monitoring, and continued professional judgment.

Ultimately, integrating technology with nursing practice can help create a safer medication-administration process. A well-designed digital verification system does not replace the nurse; instead, it provides another layer of support so nurses can identify discrepancies, document care accurately, and maintain a strong focus on patient safety.

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